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5.1 s-Block Elements:

The elements belonging to the group 1 and 2 in the modern periodic table are called s-block elements. The elements belonging to these two groups are commonly known as alkali and alkaline earth metals respectively. In this unit, we study their properties, uses, important compounds and biological importance.

5.2 Alkali metals:

The word "alkali" is derived from the word al-qalīy meaning the plant ashes, referring to the original source of alkaline substances. A water-extract of burnt plant ashes, called potash contain mainly potassium carbonate. Alkali metal group consists of the elements: lithium, sodium, potassium, rubidium, caesium and francium. They are all metals, generally soft and highly reactive. They form oxides and hydroxides and these compounds are basic in nature.

5.2.1 General characteristics of alkali metals:

Alkali metals are highly reactive and are found in nature only as compounds. Rubidium and caesium are found associated in minute quantities with minerals of other alkali metals. Francium is radioactive and does not occur appreciably in nature. Francium is highly radioactive; its longest-lived isotope has a half-life of only 21 minutes.

Table 5.1 Abundance of important alkali metals and their sources

Elements	Abundance in earth crust (%)	Relative Abundance	Mineral source
Lithium	0.0018	35	Spodumene [$LiAl(SiO_3)$]
Sodium	2.27	7	Rock Salt [NaCl]
Potassium	1.84	8	Sylvite [KCl]
Rubidium	0.0078	23	No convenient Source (obtained as by product of lithium processing)
Cesium	0.00026	46	

Figure 5.1 Alkali metals Li, Na and K stored under oil